An auto-homotopy exists as soon as one self-path is central and every point is merely connected to the basepoint

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'lem:autohtpy' (source lines 77-85); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The three item labels item:autohtpy1..3 are defined inside the retained statement and are used inside the retained proof, so they need no external entry.

Editorial note. The retained counter values make the counted result print with its original book number, and the chapter and section counters are set so that the retained context (where present) prints with its own original number as well.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: existence of an auto-homotopy from a central self-path (author label lem:autohtpy).

Allowed standard prerequisites (editorial).

- Propositional truncation. brck X is the propositional truncation of X, and a goal that is a mere proposition may be proved by assuming an element of X.. It is the technique applied at source lines 89, 103 and 113.
- Unique choice. To construct an object with a propositional existence statement one may instead characterise it uniquely and prove that the characterisation is inhabited.. It is the technique named at source lines 97-99 that leads to the auxiliary type B(x).
- Transport in identity and function types. Transport along a path in an identity type, and in a function type, reduces to componentwise statements.. It is used at source line 107 to reduce the identification of h with h' to a pointwise equation.

Source reference. Original section 3.9: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1717–1717. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the

counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original section 2.11: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1627–1627. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original section 2.9: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1628–1628. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Lemma 4.1.2. *Suppose we have a type A with $a : A$ and $q : a = a$ such that*

(i) The type $a = a$ is a set.
(ii) For all $x : A$ we have $\|a = x\|$.
(iii) For all $p : a = a$ we have $p \cdot q = q \cdot p$.

Then there exists $f : \prod_{(x:A)}(x = x)$ with $f(a) = q$.

Proof. Let $g : \prod_{(x:A)}\|a = x\|$ be as given by (ii). First we observe that each type $x =_A y$ is a set. For since being a set is a mere proposition, we may apply the induction principle of propositional truncation, and assume that $g(x) = |p|$ and $g(y) = |p'|$ for $p : a = x$ and $p' : a = y$. In this case, composing with p and ${p'}^{-1}$ yields an equivalence $(x = y) \simeq (a = a)$. But $(a = a)$ is a set by (i), so $(x = y)$ is also a set.

Now, we would like to define f by assigning to each x the path $g(x)^{-1} \cdot q \cdot g(x)$, but this does not work because $g(x)$ does not inhabit $a = x$ but rather $\|a = x\|$, and the type $(x = x)$ may not be a mere proposition, so we cannot use induction on propositional truncation. Instead we can apply the technique mentioned in original section 3.9: we characterize uniquely the object we wish to construct. Let us define, for each $x : A$, the type

$$B(x) :\equiv \sum_{(r:x=x)} \prod_{(s:a=x)} (r = s^{-1} \cdot q \cdot s).$$

We claim that $B(x)$ is a mere proposition for each $x : A$. Since this claim is itself a mere proposition, we may again apply induction on truncation and assume that $g(x) = |p|$ for some $p : a = x$. Now suppose given (r, h) and (r', h') in $B(x)$; then we have

$$h(p) \cdot h'(p)^{-1} : r = r'.$$

It remains to show that h is identified with h' when transported along this equality, which by transport in identity types and function types (original sections 2.11 and 2.9), reduces to showing

$$h(s) = h(p) \cdot h'(p)^{-1} \cdot h'(s)$$

for any $s : a = x$. But each side of this is an equality between elements of $(x = x)$, so it follows from our above observation that $(x = x)$ is a set.

Thus, each $B(x)$ is a mere proposition; we claim that $\prod_{(x:A)} B(x)$. Given $x : A$, we may now invoke the induction principle of propositional truncation to assume that $g(x) = |p|$ for $p : a = x$. We define $r :\equiv p^{-1} \cdot q \cdot p$; to inhabit $B(x)$ it remains to show that for any $s : a = x$ we have $r = s^{-1} \cdot q \cdot s$. Manipulating paths, this reduces to showing that $q \cdot (p \cdot s^{-1}) = (p \cdot s^{-1}) \cdot q$. But this is just an instance of (iii). □